

Science

General Science
Natural History
Nature Notebook
Nature Walks & Scouting





About the Course

In Level 1, learners are developing familiarity and friendship with Creation through the most accessible Things, simple text, and basic concepts.

This course includes the following topic(s): General Science: Grade 1, Natural History: Grade 1, Nature Notebook: Grade 1, Nature Walks & Scouting: Grades 1-8

About General Science: Grade 1

Students explore habitats primarily through practical observation and 'meet' some less familiar creatures from those habitats, such as bats, spiders, and seeds, through picture books. Coordinating afternoon activities are provided in Outdoor Work.

About Natural History: Grade 1

Students learn to observe a variety of creatures through their Nature Reader, including insects, birds, plants, and spiders. Coordinating afternoon activities are provided in Outdoor Work.

About Nature Notebook: Grade 1

Students spend time at least once each week recording their observations. Prompts that coordinate with curricular content can be found in Outdoor Work, if desired. This digital resource is provided with all science lessons.

About Nature Walks & Scouting: Grades 1-8

Students spend time once each week on a longer excursion. Prompts can be found in Outdoor Work, if desired. This digital resource is provided with all science lessons.

Science: Grade 1

The Big Picture:

To accomplish the goal of supporting a relationship with the Things of the Universe, a Mason science program consists of nature lore, natural history, and general science. Nature immersion, inquiry, community connection, and supportive literature are woven into each of these three parts. Form 1 learners in Grades 1-3 are at the beginning of this relationship: developing familiarity and friendship with Creation. Their sense of community is primarily focused on their immediate environment at this stage, but they are curious about others. The familiarity with foundational knowledge grown and nurtured in Form 1 prepares them for greater curiosity and personal interest in Form 2.

Nature lore is timeless knowledge that is passed through a community, much like a grandmother passes on how to make that special bread when the dough just 'feels right.' Like Mason, we strive to pass on this knowledge primarily through outdoor work. Group nature walks, seasonal readings, and topics in scouting are provided as an Outdoor Work resource in the Quick Links. If desired, literature suggestions to support lore can be found in the Community Read Alouds resource (in Literature Grades 1-3).



Placement & Combining Tips

Science: Grade 1

For first-grade students or those still growing their attention span or interest in books. If combining grades is desirable, it is recommended that the combined grades begin at the appropriate level and move forward together.

Nature Notebook: Grade 1

Learners may be combined and share prompts or follow their own interest. Support through varied media, scribing observations, or notebooking in tandem.

Nature Walks & Scouting: Grades 1-8

Learners may be combined and share prompts or follow the plan of their local scouting troop or natural history club. Support accessibility and sensory development as appropriate.



Scheduling

GRADE	SCHEDULE INFO.	BOOKS
1	General Science: Grade 1 1 time/week 15 min	A Fruit is a Suitcase for Seeds A House Spider's Life Over and Under the Pond Up in the Garden and Down in the Dirt Winter Sleep: A Hibernation Story Bat Loves the Night
1	Natural History: Grade 1 1 time/week 10 min	Walks with a Naturalist: Living Nature Reader Volume 1
1	Nature Notebook: Grade 1 1+ time/week 10 min+	
1-8	Nature Walks & Scouting: Grades 1-8 1 time/week 30 min+	

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Grade 1				
Natural History: Grade 1 Nature Walks & Scouting: Grades 1-8	Nature Notebook: Grade 1		General Science: Grade 1	



Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Science: Grade 1

- ☐ Obtain any supplies indicated on the science or grade-level supply lists.
- ☐ Download any apps and shortcut any desired links.

Nature Lore:

- ☐ Bookmark your Outdoor Work Quick Link, so that you have it available on your weekly outing. Outdoor Work is generally flexible for your location and season and can be moved around in the schedule to incorporate or substitute Natural History Club outings.
- ☐ Print or bookmark grade-specific nature notebook suggestions to support natural history and general science. Notebooking can be done on a walk, during occupations, or as a field trip, as appropriate.

Special Topics & Field Trips

Science: Grade 1

- ☐ Learn a few of your local species on these special topics using the internet, field guides, or your local nature preserve, conservancy, or state/national park.
- ☐ Note whether there are particular locations nearby for the student to easily notice the special topics on walks or during afternoon occupations.

General Science: Grade 1

- Term 1: Insects, flowers, and bats, evenings, and a pond
- Term 2: Seeds and animal homes, evenings, and a pond
- Term 3: Spiders, birds, and flowers, evenings, and a pond

Natural History: Grade 1

- Term 1: insects, spiders, and birds
- Term 2: wild seeds and birds
- Term 3: insects, spiders, and tree buds

Term Prep & Teacher Tips

Science: Grade 1

Object lesson prompts are woven into the lessons as appropriate, but teachers are encouraged to read corresponding Handbook of Nature Study (HoNS) selections each term:

General Science: Grade 1

- ☐ Term 1: Bats p.241-244, in addition to ants p.369-374 and bees p.384-395
- ☐ Term 2: Bird feathers p.29-31, and/or attracting birds p.43-44, in addition to squirrels p.233-237
- ☐ Term 3: Spiders p.435-447 and/or choose one of your garden flowers on p.546-590

☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:

- 12-18" piece of substrate (recycled cardboard, wood scrap, foam core)
- found nature objects
- various fruits, such as pit fruit (cherry, plum, peach, apricot, avocado), many-seeded fruit (lemon, melon, apple, pear), many-many seeds (banana, kiwi, berry), pea pod, ear of corn (Term 2)
- pot/planter with soil or place in the ground with trowel to dig
- any seedlings or flower seeds (Term 3)
- optional: craft paint

Natural History: Grade 1

- ☐ Term 1: ants p.369-374, bees p.384-395
- ☐ Term 2: squirrels p.233-237, 1-2 species of birds from p.50-141
- ☐ Term 3: lightning bugs p.367-368, spiders p.435-446

Reminders

General Science: Grade 1

- ☐ Fresh fruits are required for Term 2 Week 8. Make a note on your calendar to obtain these at the appropriate time.



Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

Science: Grade 1

General Science: Grade 1



A Fruit is a Suitcase for Seeds



A House Spider's Life



Over and Under the Pond



Up in the Garden and Down in the Dirt



Winter Sleep: A Hibernation Story



Bat Loves the Night

Natural History: Grade 1



Walks with a Naturalist: Living Nature Reader Volume 1



Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

∞ [View Supply List Details](#)

General Science: Grade 1



Crayola Model Magic Dough



Household Items - General Science: Grade 1



Wild Bird Seed



Bird Feeder

Nature Walks & Scouting: Grades 1-8



Wild Bird Seed



Bird Feeder



Hand Lens



Small Collection Containers



Small Basin



Dip Net



Student Microscope



Quick Links

Science: Grade 1

∞ [Outdoor Work](#)

∞ [Seek app from iNaturalist](#)

∞ [SkyView Lite for Android](#)

∞ [SkyView Lite for iOS](#)

[Click THIS text or scan the QR code for links.](#)



Science: Grade 1

How To Teach



Introduce

- Begin each lesson with learners' existing knowledge. If the book or activity is new or unfamiliar, then look at the title, a picture, or guidance in the lesson plan to help discuss what students think, drawing on previous experience. If continuing or revisiting a topic or activity, then recap. The lesson plans often help with this.
- Some learners may benefit from using pictures or looking back briefly.
- Allow them time to share any concerns and come alongside, as needed.



Complete

- Read, or do, as instructed in the lessons, noting any Teacher Tips provided. Learners should always have their nature journal or some other notebook/paper available in case they need to draw or diagram during the lesson.
- When reading to pre-readers, point to the words as you read, even if it seems like they aren't paying attention.
- If learners do not understand a word or concept, do not worry. Try to show them with a picture or connect the idea to something they have seen in real life. They are learning much by the way and will likely build understanding over the term, the year, or beyond.



Narrate

- Process the ideas of the lesson by retelling the events in sequence, describing a scene or setting, explaining a concept, etc.
- Learners may use words, pictures, Legos, dolls, etc., to process and convey ideas in the manner most natural to them.
- Teachers may take turns to model.



Discuss

- Consider together any thoughts, confusion, or concerns about the passage.
- Questions/topics for further discussion are often provided in the lesson plans to help. There are no right or wrong answers to these. If teachers want to keep ideas active in the mind, these can also be used at other times to keep the ideas in the working memory.



Connect

- Follow any extra links, examine any sidebars in the text, look at pictures, etc., depending on learner interest.
- These can also be viewed as alternative ways to engage.

Term 1

WEEK 1 10m Natural History: Grade 1 - Lesson 1

A Sense of Fall

 Materials: Walks with a Naturalist 1

PREP: Read Teacher Tip

→ INTRO

Have you ever gone on a walk with a naturalist or park ranger? If so, what was that like? These are people who spend a LOT of time getting to know nature. This book has been written especially for YOU by a naturalist! Today, she is going to introduce herself and help us start to think about the upcoming season. She'll take us on her walks all year long!

→ READ, NARRATE, & DISCUSS

p.xi-3 "Hello, readers!" - "go outside!"

Read the definitions and notes at the end of each reading, as desired.

- When you think about the fall season, what do you think of? What do you look forward to? Have you noticed any of these signs of fall yet?

→ SUPPLEMENTAL

∞ Video Link: Fall Colors in Texas (2:08)

∞ Video Link: Wood Thrush Song (1:35)

∞ Video Link: Senses Nature Walk (6:34)

★ TEACHER TIP

Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 1 15m General Science: Grade 1 - Lesson 1

Listening to the Chorus Activity

 Materials: Nature Journal

PREP: Read Teacher Tip.

→ INTRO

When we observe, we can use all our senses, but we often rely on sight more than our other senses. Use your time today to sit and just listen to the sounds of nature.

→ LISTEN

Find a place to sit where you can hear the sounds of nature. Sit in a comfortable position, maybe with your eyes closed. Listen for about 7-10 minutes.

→ NARRATE & DISCUSS

Use a Nature Journal to record what the student heard.

- What sounds were the loudest? What sounds did you hear most often? Were there any that you had to listen very closely to hear? Do you know who you heard? Do you know where they live?

★ TEACHER TIP

If students seem anxious or unsure about their Nature Journals, teachers can encourage them by modeling, showing them many different examples, journaling in tandem, or serving as a scribe.

WEEK 2 10m Natural History: Grade 1 - Lesson 2

Shake a Bush

 Materials: Walks with a Naturalist 1

PREP: Read Teacher Tip

→ INTRO

Look at the picture on p.4. Have you ever seen a creature like this before? What kinds of 'bugs' have you noticed? Briefly share in word or picture.

★ TEACHER TIP

It can sometimes be unnerving for students not to get an immediate answer to a question. Encourage them to sit with big questions, like the one in today's suggested discussion, while they continue to make observations.



Term 1

→ READ, NARRATE, & DISCUSS
p.4-6 "One of my" - "you caught them."

- What do you think makes a leafhopper a leafhopper? What does it do? How is it made?

→ SUPPLEMENTAL
∞ Video Link: Jumping Hoppers in Slow Motion (1:44-3:23)

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital!
Resources in Quick Links.

WEEK 2 15m General Science: Grade 1 - Lesson 2

Pond Habitat

Materials: Over and Under the Pond

PREP: Read Teacher Tip.

→ INTRO
Look at the picture on the cover. The creatures that we see and hear outside all have homes just like us! These homes are called habitats. Let's read a book today about one habitat, called a pond. If you have been to a pond, tell about or draw one thing that stood out to you. Did you ever think about a pond as someone's home?

→ READ, NARRATE, & DISCUSS
Over and Under the Pond (whole book)
• If you have visited a pond before, who do you remember seeing there? If not, what or who do you hope to see when we visit?

★ TEACHER TIP
Let students enjoy the artwork in this picture book. If you need to take a break and finish it another time, that is fine.

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

WEEK 3 10m Natural History: Grade 1 - Lesson 3

Ants in My Plants

Materials: Walks with a Naturalist 1

PREP: Read Teacher Tip

→ INTRO
Draw a picture or diagram of an ant or other insect (or a partner might diagram alongside a verbal drawing). What parts do you recall?

→ READ, NARRATE, & DISCUSS
p.8-11 "In 'Shake'" - "and differences!"
• Compare the ant on p.9 to the leafhopper on p.5. How are they similar? How are they different?
• Add to the drawing you did before the reading or draw a new one. Can you remember the parts of an insect? (3 body segments, 6 legs)
• Have you noticed insects outside? What were they doing? Where were they? What made that place a good home?

→ SUPPLEMENTAL
∞ Video Link: All the Life Stages of an Ant (1:34)
∞ Video Link: Flying Ants Up Close (5:22)

★ TEACHER TIP
If students need support drawing or diagramming, the teacher may do any activities with them in tandem or act as their scribe.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital!
Resources in Quick Links.

WEEK 3 15m General Science: Grade 1 - Lesson 3

Same Creatures, Different Homes Activity

Materials: Nature Journal

→ INTRO
Some creatures live in only one particular habitat, while others live in many different



Term 1

habitats. Today let's notice creatures that live in many different habitats.

→ OBSERVE

Spend about 10 minutes noticing creatures around home/school that also live at the pond. These could be insects, squirrels, and even trees or grasses.

→ NARRATE & DISCUSS

Use a Nature Journal to record what the student noticed.

- Did you notice any creatures that live both at home/school AND at the pond? If so, how do you think their lives are different in these different habitats? If not, why do you suppose not?

WEEK 4 10m Natural History: Grade 1 - Lesson 4

Splooting and Anting

Materials: Walks with a Naturalist 1

→ INTRO

Recall what we read. Tell about or draw/diagram an insect. What parts do you recall? Ants live with other creatures in nature. Do you think you would like to live with ants? Why or why not? What do you think the bird is doing in the picture on p.13?

→ READ, NARRATE, & DISCUSS

p.13-15 "I love" - "around them."

- Have you ever noticed animals doing funny or curious things?

→ SUPPLEMENTAL

∞ Video Link: Sploot Season (0:55)

∞ Video Link: Birds Bathing on an Ant Hill (2:22)

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital!

Resources in Quick Links.

WEEK 4 15m General Science: Grade 1 - Lesson 4

Looking for Ants Activity

Materials: Nature Journal

PREP: Read Teacher Tip.

→ INTRO

Today, you will look for ants. Some good places to look are under small rocks, under loose bark, and in cracks.

→ OBSERVE

Use a stick to safely lift rocks and bark. If you find ants, take some time to observe them. If you find someone else that you want to observe, you may take the time to do so. Spend about 10 minutes looking for ants.

→ NARRATE & DISCUSS

Use a Nature Journal to record what the student noticed.

- Did you find ants? Some other interesting creature? Tell about their habitat: where is it? What is around? What was the creature doing? Do they have a nest? If so, how do you know? Did you notice eggs, larvae, or pupae? Did they carry food? Do you notice how they entered or left their home?

★ TEACHER TIP

This week or next is a great opportunity to do an object lesson during the science activity. Questions are embedded in the lesson to help, as needed. If preferred, teachers can instead refer to the Handbook of Nature Study or allow questions to flow naturally between student and teacher.



Term 1

WEEK 5 10m Natural History: Grade 1 - Lesson 5

Hot Ants

Materials: Walks with a Naturalist 1

PREP: Read Teacher Tip

→ INTRO

Tell about our last reading. What do you remember about insects? Briefly share in word or picture.

→ READ, NARRATE, & DISCUSS

p.16-19 "When Mike" - "live in?"

- Have you noticed an ant's home before? Describe what you noticed.
- Did you know that insects could be farmers? What kinds of work have you seen ants doing?

→ SUPPLEMENTAL

∞ Video Link: Ant Farm Time Lapse (3:20)

∞ Video Link: Ants Farming Aphids (2:01)

★ TEACHER TIP

Notice that any prompt to recall or think about can be done verbally, in picture, or in any other way that suits the student(s). Apply this flexibility to all lessons.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 5 15m General Science: Grade 1 - Lesson 5

Looking for Ants Activity

Materials: Nature Journal

→ INTRO

Look for ants again this week or return to visit your ants again, if you found some last week.

→ OBSERVE

Is anything different this week? If you find an ant, carefully take a closer look at its body, being careful not to disturb a nest; you might place an ant in a container temporarily and look at it with a magnifier. How many parts of the body can you count? How many legs? From which parts do the legs come? How many antennae does it have? How does it use its antennae? Spend about 10 minutes looking for/watching ants.

→ NARRATE & DISCUSS

Use a Nature Journal to record what the student noticed.

- Tell about what you found or what you recorded.

WEEK 6 10m Natural History: Grade 1 - Lesson 6

Counting Bumbles

Materials: Walks with a Naturalist 1

PREP: Read Teacher Tip

→ INTRO

What do you remember about our last reading? Look at the picture on p.20. Do you know who that is? We're going to read about another insect today and find out why he's in that tube. If it's an insect, what body parts does that mean it will have? (6 legs and 3 body segments: head, thorax, abdomen)

→ READ, NARRATE, & DISCUSS

p.20-24 "Did you know" - "nature journal."

★ TEACHER TIP

Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your

Term 1

• Bumble bees are insects just like ants and leafhoppers. Looking at the picture on p.22, how is the bumble bee different from ants and leafhoppers? → SUPPLEMENTAL ∞ Video Link: Bumble bees in slow motion (watch as much as you want, trying to notice the legs, body segments, and any other interesting features or behaviors)	student(s). • OUTDOOR WORK Weekly walks and daily outdoor time are vital! Resources in Quick Links. Resources in Quick Links.
WEEK 6 □ 15m General Science: Grade 1 - Lesson 6 <i>Noticing Bees Activity</i> 📄 Materials: Nature Journal PREP: Read Teacher Tip. → INTRO Look for other insects this week, especially bees. → OBSERVE Spend about 10 minutes noticing insects. What are they doing? What parts of the body can you see? How would you describe the habitat? → NARRATE & DISCUSS Use a Nature Journal to record what the student noticed. • How is this bee (or other insect) similar to and different from the ant?	★ TEACHER TIP The bees will begin slowing down toward the end of fall. Notice if this is happening, as you might get to see one up close and do an object lesson!
WEEK 7 □ 10m Natural History: Grade 1 - Lesson 7 <i>Scary Spider? No Way!</i> 📄 Materials: Walks with a Naturalist 1 PREP: Read Teacher Tip → INTRO Tell me what we've been reading about. Look at the picture on p.25. Does this creature have three body segments and six legs, like the ant and bumble bee? How many body segments and how many legs does the spider have? Spiders are not insects! Have you ever watched a spider or seen a spider web before? Tell about it. → READ, NARRATE, & DISCUSS p.25-28 "I used to be" - "look like?" • What do you want to notice about the next spider web you find? Where do you think an orb weaver would make its home? → SUPPLEMENTAL ∞ Video Link: Orb Weavers (1:22) ∞ Video Link: Amazing Spider with Huge Web (3:15)	★ TEACHER TIP It can sometimes be unnerving for students not to get an immediate answer to a question. Encourage them to sit with big questions, like the one in today's suggested discussion, while they continue to make observations. • OUTDOOR WORK Weekly walks and daily outdoor time are vital! Resources in Quick Links.
WEEK 7 □ 15m General Science: Grade 1 - Lesson 7 <i>Noticing Bees Activity</i> 📄 Materials: Nature Journal → INTRO Look for bees again this week. → OBSERVE Is anything different this week? Do you find the same type of bee, a different type,	• NATURE NOTEBOOK Prompt for General Science in Outdoor Work Quick Link.



Term 1

or none at all? Some other type of insect? Spend about 10 minutes noticing.

→ NARRATE & DISCUSS

Use a Nature Journal to record what the student noticed.

- Tell about what you found or what you recorded.

WEEK 8 10m Natural History: Grade 1 - Lesson 8

Cute Spiders?

Materials: Walks with a Naturalist 1

PREP: Read Teacher Tip

→ INTRO

Remind me what we read about last. How was the garden spider we met last week different from the insects we met earlier in the book? Let's meet another spider today. Compare the pictures on p.27, 30. How are the spiders similar? How are they different?

→ READ, NARRATE, & DISCUSS

p.30-33 "Do you think" - "your mind!"

- Did you notice how some jumping spiders make their home in a leaf? Where do you think they might live outside? That place where they find food and shelter is their habitat!

→ SUPPLEMENTAL

∞ Video Link: Spider with Super Powers (4:00)

★ TEACHER NOTE

The spider in the supplemental video catches prey, which some students might find disturbing. Teachers should preview or skip this one if they have concerns.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 8 15m General Science: Grade 1 - Lesson 8

Meeting an Evening Creature

Materials: Bat Loves the Night

PREP: Read Teacher Tip.

→ INTRO

One creature that you may have noticed in the evening is the bat. Have you ever seen a bat? If so, what did you notice about them? Briefly describe in words or a picture. Bats need very particular types of habitats. They are also very important helpers to many other creatures. Let's read and find out more.

→ READ, NARRATE, & DISCUSS

Bat Loves the Night (whole book)

- What kinds of things do you think the bat needs in its habitat?

★ TEACHER TIP

Let students enjoy the artwork in this picture book. If you need to take a break and finish it another time, that is fine.

• NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.

WEEK 9 10m Natural History: Grade 1 - Lesson 9

Chimney Swift Tornado

Materials: Walks with a Naturalist 1

→ INTRO

What are some of the creatures we've read about this term? Where do they live and find habitat (a place with food and shelter) in the world around us? Do they ever come into your home? Do we share the same habitat as insects and spiders or are their homes separate from ours? Today we'll read about a bird whose habitat is definitely part of someone else's! Look at the picture on p.34.

→ READ, NARRATE, & DISCUSS

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.



Term 1

p.34-36 "One evening" - "they roost?"

- If our homes are part of the natural world, then the creatures in nature are our neighbors! Tell how you care for your neighbors, human or animal.

→ SUPPLEMENTAL

- ∞ Video Link: Chimney Swifts (0:45)
- ∞ Video Link: Portland Swift Tornado (watch as much as you like)

WEEK 9 15m General Science: Grade 1 - Lesson 9

Noticing Sound Activity

Materials: Nature Journal

→ INTRO

It can be hard to understand exactly how a bat uses echolocation because we don't share that experience, but we can think about how different sounds sound different.

→ ACTIVITY

Choose three different objects, such as a penny, a bead, and a small ball of paper. Drop these three objects onto three different surfaces to hear the sound that bounces back to your ear, such as a metal pan, the floor, and a countertop or desk. Do you think you could learn to tell these apart if you couldn't see them? Spend about 10 minutes listening and describing what you hear.

→ NARRATE & DISCUSS

- Use a Nature Journal to record what the student noticed.
- Tell about what you found or what you recorded.

WEEK 10 10m Natural History: Grade 1 - Lesson 10

Yard Elk

Materials: Walks with a Naturalist 1

→ INTRO

Tell what you remember about the chimney swift. Have you noticed any bird neighbors this week? What were they doing? Have you noticed signs that they may be getting ready for a change in season? Today our naturalist friend tells us about another animal neighbor that she is getting to know.

→ READ, NARRATE, & DISCUSS

p.38-40 "Today, a herd" - "for winter?"

- How do you think animals that live outside all year round survive in very cold seasons?

→ SUPPLEMENTAL

- ∞ Video Link: Elk in the Ozarks (2:19)
- ∞ Video Link: Bighorn Sheep, another ruminant in a different habitat (3:20)
- ∞ Video Link: White-tailed Deer, another ruminant in a different habitat (2:32)

WEEK 10 15m General Science: Grade 1 - Lesson 10

Modeling a Pond Habitat Activity

Materials: Clay, 12-18" piece of substrate, e.g., cardboard, wood scrap, foam core, found nature objects, optional: craft paint

PREP: Read Teacher Tip.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

★ TEACHER TIP

Students may need to work on their models in the afternoon if they need extra time.

Term 1

→ INTRO

What are some things you have learned about ponds this term? Let's use the last 2 weeks of the term to create a model of a pond habitat. You will use clay to form your pond, paint if desired, and then you can use found nature objects to add the plants and animals that live there.

→ ACTIVITY

∞ Website Link: Salt Dough Recipe (as an alternative to clay)

WEEK 11 10m Natural History: Grade 1 - Lesson 11

Catch Up

→ CATCH UP

Use this time to catch up on lessons, watch some videos that you didn't have time for, or discuss some interesting ideas over a snack. Depending on your discussions, these MIGHT include:

- What do you remember about ants?
- How are bees similar to and different from ants?
- How are spiders different from insects?
- Other creatures as our neighbors.
- Signs of the season with all our senses and signs that animals are preparing for that season.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 11 15m General Science: Grade 1 - Lesson 11

Modeling a Pond Habitat Activity

 Materials: Clay, found nature objects, optional: craft paint

→ ACTIVITY

Complete the pond habitat model.

∞ Website Link: Salt Dough Recipe (as an alternative to clay)

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

WEEK 12 10m Natural History: Grade 1 - Lesson 12

Exam Week

→ EXAMS

Answer question(s) related to course.

Questions will come from:

Walks with a Naturalist 1

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 12 15m General Science: Grade 1 - Lesson 12

Exam Week

→ EXAMS

Answer question(s) related to course.

Questions will come from:

Over and Under the Pond

Term 2

WEEK 1 10m Natural History: Grade 1 - Lesson 13

Sticky Seeds

 Materials: Walks with a Naturalist 1

PREP: Read Teacher Tip

→ INTRO

Look at the picture on p.41. Did you ever find stuff stuck to you after you went out to play or walk? What did it look like? Where did it come from?

→ READ, NARRATE, & DISCUSS

p.41-42 "Sometimes Mike" - "closer look."

Read the notes and definitions at the end of each reading, as desired.

- Have you ever seen the seeds of a wild plant? If so, briefly tell about it.
- Tell a story about where a seed goes and how it finds its new home after it sticks to an animal.

→ SUPPLEMENTAL

∞ Video Link: Showy Tick Trefoil (1:07)

★ TEACHER TIP

Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 1 15m General Science: Grade 1 - Lesson 13

Habitats in Winter

 Materials: Winter Sleep

PREP: Read Teacher Tip.

→ INTRO

Does your home or the way you live change in the winter? Look at the picture on the cover. What do you think happens to the creatures' habitats when the seasons change? How do plants and animals get ready for winter? Briefly describe in words or a picture.

→ READ, NARRATE, & DISCUSS

Winter Sleep: A Hibernation Story (whole book)

You may leave the end matter for another time based on student interest.

- Have you ever come across an animal that was sleeping for the winter? If so, what did you notice? What do you notice the non-hibernating animals doing in the winter? What happens to plants in winter?

★ TEACHER TIP

Let students enjoy the artwork in this picture book. If you need to take a break and finish it another time, that is fine.

• NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.

WEEK 2 10m Natural History: Grade 1 - Lesson 14

Pinecone Bombs

 Materials: Walks with a Naturalist 1

PREP: Read Teacher Tip

→ INTRO

Remind me about the seeds we read about last week. Look at the picture on p.44. What do you suppose the squirrel is doing on that branch?

→ READ, NARRATE, & DISCUSS

p.44-46 "Mike and I" - "seeds inside."

- Last week we met a plant that helps its seeds move by making them sticky. How did this week's plant help its seeds move? (Might mention animals eating or burying,

★ TEACHER TIP

It can sometimes be unnerving for students not to get an immediate answer to a question. Encourage them to sit with big questions, like the one in today's suggested discussion, while they continue to make observations.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.



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wind). Can you think of other ways that seeds move around to find a good home?
How is the seed helped by the way it was made?

→ SUPPLEMENTAL

- ∞ Video Link: Pine Tree Time Lapse (1:55)
- ∞ Image Link: Winged seeds of Norway Spruce

WEEK 2 15m General Science: Grade 1 - Lesson 14 *Providing for Animals in Winter Activity*

Materials: Bird feeder and/or sticks/logs/leaves

→ ACTIVITY

Before it gets too cold, put up a bird feeder (purchased or homemade) and/or build a simple stick/log/leaf pile.
∞ Website Link: Fun DIY Bird Feeders

★ TEACHER NOTE

Note that students should collect a variety of wild seeds over the next couple of weeks, as prompted in their Nature Notebook suggestions. These are for Wk 5 Lsn 50 and Wk 10 Lsn 65.

- NATURE NOTEBOOK Prompt for General Science in Outdoor Work Quick Link.

WEEK 3 10m Natural History: Grade 1 - Lesson 15 *The Bird with the Best Memory*

Materials: Walks with a Naturalist 1

→ INTRO

Let's remember the seeds we've read about so far and how they move around. Today we're going to read about a different kind of pine seed and an animal that helps this plant move around.

→ READ, NARRATE, & DISCUSS

p.48-50 "One fall morning" - "your dinner!"

- Have you noticed a bird stuffing its mouth or flying busily back and forth at the bird feeder? Maybe it was caching! Act like the bird and show what it did.
- A keystone species is a creature that many other creatures rely on. Have you noticed a tree that many other creatures rely on? Tell about it.

→ SUPPLEMENTAL

- ∞ Video Link: Clark's Nutcracker (1:35)
- ∞ Video Link: American Crow preparing to cache (1:29)
- ∞ Website Link: All About Birds, scroll down to see some different birds that cache and the different strategies they use

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 3 15m General Science: Grade 1 - Lesson 15 *Noticing Animal Activity*

Materials: Nature Journal

PREP: Read Teacher Tip.

→ INTRO

Spend about 10 minutes watching animal behavior during this time of seasonal change.

→ OBSERVE

What are the animals doing? Is anyone visiting your feeder or pile?

★ TEACHER TIP

If students seem anxious or unsure about their Nature Journals, teachers can encourage them by modeling, showing them many different examples, journaling in tandem, or serving as a scribe.

- NATURE NOTEBOOK Prompt for General Science



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→ NARRATE & DISCUSS Use a Nature Journal to record what the student noticed. • Tell about what you found or what you recorded.	in Outdoor Work Quick Link.
WEEK 4 10m Natural History: Grade 1 - Lesson 16 <i>A Bear-y Nice Tree</i> Materials: Walks with a Naturalist 1 → INTRO What kinds of seeds have we been reading about? Tell me about the animals that like to eat seeds and fruits. Today, let's read about one that Melissa observed in her yard. → READ, NARRATE, & DISCUSS p.51-53 "Mike and I" - "eat the fruit!" • What does the seed do for the plant? What does the seed do for the animals? → SUPPLEMENTAL ∞ Video Link: The Outdoor Theater (16:57). Watch as much or as little as you like; watch on increased speed if short on time.	• OUTDOOR WORK Weekly walks and daily outdoor time are vital! Resources in Quick Links.
WEEK 4 15m General Science: Grade 1 - Lesson 16 <i>Noticing Animal Activity</i> Materials: Nature Journal PREP: Read Teacher Tip. → INTRO Spend about 10 minutes watching animal behavior during this time of seasonal change. → OBSERVE What are the animals doing? Is anyone visiting your feeder or pile? → NARRATE & DISCUSS Use a Nature Journal to record what the student noticed. • What was different this week compared to last week?	★ TEACHER TIP Animals that come to your feeder or pile provide a great opportunity to do an object lesson during the science activity. • NATURE NOTEBOOK Prompt for General Science in Outdoor Work Quick Link.
WEEK 5 10m Natural History: Grade 1 - Lesson 17 <i>Salamander Night Out</i> Materials: Walks with a Naturalist 1 → INTRO What kinds of creatures have we been reading about? Look at the picture on p.55. This is a very different kind of animal! What do you notice about the salamander? → READ, NARRATE, & DISCUSS p.55-59 "It was a" - "of their own!" • Have you seen a salamander before? If so, tell about it. Does the salamander remind you of any other creature? Have you seen its cousin, the frog? → SUPPLEMENTAL ∞ Video Link: Life cycle of a Salamander (3:39)	• OUTDOOR WORK Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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WEEK 5 15m General Science: Grade 1- Lesson 17

Looking Closely at Seeds Activity

 Materials: Nature Journal

PREP: Read Teacher Tip.

→ INTRO

Choose at least one of the collected seeds to draw/paint today. Look closely at it before you begin. What do you notice? How is this seed built? Does it remind you of anything else? Does it give you an idea as to how the seed spreads for planting? Does an animal eat it? Does it hold on to an animal to ride along? Does it ride on the wind?

→ NARRATE & DISCUSS

Use a Nature Journal to record what the student noticed.

- Tell about what you recorded. Do you know the name of the plant that made this seed?

★ TEACHER TIP

This is a great opportunity to do an object lesson during the science activity.

- NATURE NOTEBOOK Prompt for General Science in Outdoor Work Quick Link.

WEEK 6 10m Natural History: Grade 1- Lesson 18

Christmas Bird Count Part 1

 Materials: Walks with a Naturalist 1

PREP: Read Teacher Tip

→ INTRO

Recall who we read about last week. Do you remember the animals we read about before that? Where do you like to go to watch animals in nature? Let's read about a place where Melissa likes to go.

→ READ, NARRATE, & DISCUSS

p.60-62 "Every year" - "natural spectacle!"

- Tell about the bird you see most at the feeder. How many different birds do you think you see at the feeder?

→ SUPPLEMENTAL

∞ Video Link: Counting the Birds (1:59)

★ TEACHER TIP

Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

- OUTDOOR WORK Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 6 15m General Science: Grade 1- Lesson 18

Noticing Animal Activity

 Materials: Nature Journal

→ INTRO

Spend about 10 minutes watching animal behavior during this time of seasonal change.

→ OBSERVE

What are the animals doing? Is anyone visiting your feeder or pile?

→ NARRATE & DISCUSS

Use a Nature Journal to record what the student noticed.

- What was different this week compared to previous weeks?

- NATURE NOTEBOOK Prompt for General Science in Outdoor Work Quick Link.



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WEEK 7 10m Natural History: Grade 1 - Lesson 19

Christmas Bird Count Part 2

Materials: Walks with a Naturalist 1

→ INTRO

What did we read about last? Did you see birds this week? Did you hear birds this week? Let's find out what other birds Melissa noticed on her Count.

→ READ, NARRATE, & DISCUSS

p.63-66 "On our Christmas" - "can you find?"

- Is there a bird song you hear often? Can you sing that song? Have you seen who it belongs to? Where do you hear it?

→ SUPPLEMENTAL

∞ Website Link: Sounds & Songs

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 7 15m General Science: Grade 1 - Lesson 19

Thinking about Seeds

Materials: A Fruit is a Suitcase for Seeds

PREP: Read Teacher Tip.

→ INTRO

What kinds of fruit do you like to eat? Do you know what kind of plant makes your fruit? Why does the plant make fruit? Look at the picture on the cover. Does your fruit have seeds?

→ READ, NARRATE, & DISCUSS

A Fruit is a Suitcase for Seeds (whole book)

- What do you think happens to seeds in nature when an animal eats the fruit? Do all plants that make seeds package them inside fruit? What about the seed you drew/painted last week? Did it have a fruit?

★ TEACHER TIP

Let students enjoy the artwork in this picture book. If you need to take a break and finish it another time, that is fine.

★ TEACHER NOTE

Students will need a variety of fruits to examine their seeds next week.

WEEK 8 10m Natural History: Grade 1 - Lesson 20

Flowers to Seeds

Materials: Walks with a Naturalist 1

→ INTRO

Remind me of what we read about last time. What do birds eat during the winter? Look at the picture on p.67. What is that bird doing?

→ READ, NARRATE, & DISCUSS

p.67-70 "Our yard" - "a winter meal?"

- Tell about a place where you saw wildflowers in the summer or fall. Have you noticed who is eating at that flower in winter?

→ SUPPLEMENTAL

∞ Video Link: Goldfinches Eating (2:59)

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.



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WEEK 8 15m General Science: Grade 1 - Lesson 20

Investigating Fruits Activity

Materials: Nature Journal, a variety of fruits with seeds

→ INTRO

Have you ever noticed the different kinds of seeds in the fruit you eat? Let's investigate today!

→ ACTIVITY

Open a pit fruit (cherry, plum, peach, apricot, avocado), a many-seeded fruit (lemon, melon, apple, pear), a many-many seeded fruit (banana, kiwi, berry), a pea pod, and an ear of corn to find the seeds. Look closely and compare them. How are they different from each other? How are they similar? When you are finished, choose one or more that you would like to plant. Wash them and lay them on a towel to dry.

→ NARRATE & DISCUSS

Use a nature journal to record what the student noticed.

- Tell about what you found or what you recorded.

WEEK 9 10m Natural History: Grade 1 - Lesson 21

The Round House

Materials: Walks with a Naturalist 1

→ INTRO

Tell me what we read about last week. Look at the picture on p.71. What do you notice? What do you think is going on in the picture?

→ READ, NARRATE, & DISCUSS

p.71-72 "I am" - "goldenrod gall fly!"

- Have you ever thought of what it would be like to be an insect? Tell a story about an insect you've seen from the insect's point-of-view.

→ SUPPLEMENTAL

∞ Video Link: What Gall! (2:58)

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 9 15m General Science: Grade 1 - Lesson 21

Noticing Animal Activity

Materials: Nature Journal

→ INTRO

Spend about 10 minutes watching animal behavior during this time of seasonal change.

→ OBSERVE

What are the animals doing? Is anyone visiting your feeder or pile?

→ NARRATE & DISCUSS

Use a Nature Journal to record what the student noticed.

- What was different this week compared to previous weeks?



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WEEK 10 10m Natural History: Grade 1 - Lesson 22

The Goldenrod Gall Fly

Materials: Website link

→ INTRO

What did we read about last? Look at the picture on p.74. What do you notice about this insect?

→ READ, NARRATE, & DISCUSS

p.74-78 "Mike and I" - "find a gall!"

- Galls are just one way that some insects raise their babies. If you have ever seen a wasp nest or bees emerging from the ground, then you've seen other homes for baby insects. Tell about any kinds of insect homes that you have seen.
- We often don't see many insects in the colder season, but if you watched either of the supplemental videos, can you see the parts that make these creatures insects? (3 body segments, 6 legs)

→ SUPPLEMENTAL

∞ Video Link: Thistle Stem Gall Fly (0:22)

∞ Video Link: Oak Apple Gall Wasps (1:54)

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 10 15m General Science: Grade 1 - Lesson 22

Looking Closely at Seeds Activity

Materials: Nature Journal

→ INTRO

Draw/paint a seed today.

→ OBSERVE

Choose another of the collected seeds to draw/paint today. Look closely at it before you begin. What do you notice?

→ NARRATE & DISCUSS

Use a Nature Journal to record what the student noticed.

- How does this compare to the seeds obtained from fruits that you eat? Do you think an animal eats this seed? If not, does the seed offer any clues as to how it might spread, so that it can grow? What is its shape? What is its texture?

WEEK 11 10m Natural History: Grade 1 - Lesson 23

Catch Up

→ CATCH UP

Use this time to catch up on lessons, watch some videos that you didn't have time for, or discuss some interesting ideas over a snack. Depending on your discussions, these MIGHT include:

- How does the shape and structure of a seed help the plant move its seed to a new home?
- How do animals help plants? How do plants help animals? (Think about squirrels, birds, bears, insects, etc.)
- Tell about salamanders.
- Tell about the different kinds of animal homes you can think of.
- What kinds of birds have you seen or heard?

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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WEEK 11	15m General Science: Grade 1 - Lesson 23 Planting a Seed Activity Materials: Pot, soil, seed, water → ACTIVITY Plant one or more of the fruit seeds that you saved. Plant it as deep as the seed is wide and give it just enough water to dampen the soil. You will need to continue to keep the soil moist, but not wet, over break. When your seed germinates, place the pot near a window where it will receive sunlight.	
WEEK 12	10m Natural History: Grade 1 - Lesson 24 Exam Week → EXAMS Answer question(s) related to course. Questions will come from: Walks with a Naturalist 1	• OUTDOOR WORK Weekly walks and daily outdoor time are vital! Resources in Quick Links.
WEEK 12	15m General Science: Grade 1 - Lesson 24 Exam Week → EXAMS Answer question(s) related to course. Questions will come from: A Fruit is a Suitcase for Seeds	



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WEEK 1 10m Natural History: Grade 1 - Lesson 25

Opening Buds

Materials: Walks with a Naturalist 1

PREP: Read Teacher Tip

→ INTRO

Look at the picture on p.79. Did you know that trees have buds just like garden plants? What do you notice about this bud?

→ READ, NARRATE, & DISCUSS

p.79-82 "Every year" - "start to open!"

Read the notes and definitions at the end of each reading, as desired.

- What signs of spring have you noticed?

→ SUPPLEMENTAL

∞ Video Link: Oak Leaf Budding (0:32)

∞ Video Link: Beech Twig Flowers and Leaves Budding (1:22)

★ TEACHER TIP

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• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 1 15m General Science: Grade 1 - Lesson 25

Bird Watching Activity

Materials: Nature Journal

→ INTRO

Today we will notice the birds.

→ OBSERVE

Spend about 10 minutes bird watching. What colors, markings, and shapes do you notice? What are they doing?

→ NARRATE & DISCUSS

Use a Nature Journal to record what the student noticed.

- Do you notice anyone already familiar to you? Do you notice anyone new?

• NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.

WEEK 2 10m Natural History: Grade 1 - Lesson 26

Spring Babies

Materials: Walks with a Naturalist 1

→ INTRO

What do you remember about buds opening from the last reading? Baby animals also emerge in the spring. Have you ever watched a baby animal?

→ READ, NARRATE, & DISCUSS

p.83-85 "In Yellowstone" - "on a farm."

- Have you spotted baby birds or other nature babies outside? What were they doing?

→ SUPPLEMENTAL

∞ Video Link: Baby Animals of Yellowstone (4:08)

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.



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WEEK 2 15m General Science: Grade 1 - Lesson 26

Garden Habitat

Materials: Up in the Garden and Down in the Dirt

PREP: Read Teacher Tip.

→ INTRO

Look at the picture on the cover. Have you ever thought about your yard or garden as a habitat? Do you already know some creatures who live there?

→ READ, NARRATE, & DISCUSS

Up in the Garden and Down in the Dirt (whole book)

- Have you seen or heard any of the creatures in the story before? Are there any that you think would be interesting?

★ TEACHER TIP

Let students enjoy the artwork in this picture book. If you need to take a break and finish it another time, that is fine.

WEEK 3 10m Natural History: Grade 1 - Lesson 27

A Yard Bouquet

Materials: Walks with a Naturalist 1

→ INTRO

Tell me about the signs of new life that we've been reading about (buds and baby animals). New flowers are another sign of spring. Have you ever made a bouquet of wildflowers? Tell about your favorite wildflowers. Look at the picture of Melissa's favorite wildflowers on p.87. Do you know any of them?

→ READ, NARRATE, & DISCUSS

p.87-91 "In spring" - "look closely?"

- Tell about your favorite wildflower. When and where do you see it?

→ SUPPLEMENTAL

∞ Video Link: California Wildflowers (0:49)

∞ Video Link: New York Wildflowers (0:59)

∞ Video Link: Desert Wildflowers (3:58)

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 3 15m General Science: Grade 1 - Lesson 27

Preparing the Garden Activity

Materials: Pot with soil or patch of ground, seeds or seedlings, water

PREP: Read Teacher Tip.

→ ACTIVITY

Help prepare a small garden - even a single pot can be a habitat! Plant some seeds/seedlings, or transplant the one that sprouted last term. Remember that you will need to check on it regularly to pull a weed or provide water, as needed.

★ TEACHER TIP

It can help to set a reminder in your calendar while students are forming the habit of caring for their garden.

WEEK 4 10m Natural History: Grade 1 - Lesson 28

Rolling Logs

Materials: Walks with a Naturalist 1

→ INTRO

Remind me of what we read about last time. Have you ever thought of a dead tree as a sign of life? When you've been outside, have you ever come across a log on the ground? If so, tell what you noticed. Maybe you noticed someone like in the picture

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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on p.92. What do you notice about this creature?

→ READ, NARRATE, & DISCUSS

p.92-96 "In our North" - "where you live!"

Be sure to read the safety note on p.97.

- How many different creatures did Melissa see under logs?
- What would happen if you rolled the log away from you instead of toward you while looking over?

∞ Video Link: Life Under a Rotting Log (5:29), similar to the one linked in your text

∞ Video Link: Earwig wings are origami-like (2:41)

WEEK 4 15m General Science: Grade 1 - Lesson 28

Bird Watching Activity

 Materials: Nature Journal

→ INTRO

Today we will notice the birds.

→ OBSERVE

Spend about 10 minutes bird watching. What colors, markings, and shapes do you notice? What are they doing?

→ NARRATE & DISCUSS

Use a Nature Journal to record what the student noticed.

- Do you notice anyone already familiar to you? Do you notice anyone new?

WEEK 5 10m Natural History: Grade 1 - Lesson 29

Tree Holes

 Materials: Walks with a Naturalist 1

PREP: Read Teacher Tip

→ INTRO

Tell me about the log habitat we read about last week. Have you ever noticed a living tree with holes in it? If so, briefly tell about what you noticed. Today we're going to read about another way that trees provide a place for creatures to live and grow.

→ READ, NARRATE, & DISCUSS

p.98-101 "Have you ever" - "make their nests."

- Tell about the different creatures you have seen in a tree.

→ SUPPLEMENTAL

∞ Video Link: Mystery of the Tiny Tree Holes (4:17)

★ TEACHER TIP

Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 5 15m General Science: Grade 1 - Lesson 29

Critter Investigation Activity

 Materials: Nature Journal, garden trowel, piece of ground to dig

→ INTRO

We have spent time noticing creatures in the sky and on the surface of the earth, but what about under the surface? Does anyone live there?

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→ OBSERVE

Spend about 10 minutes digging in the dirt or leaf litter to see who is busy under the surface. If you see a pill bug, what does it do when you pick it up? If you see a worm, how does it feel in your hand? Who is the first visitor up above the surface?

→ NARRATE & DISCUSS

Use a Nature Journal to record what the student noticed.
• Tell about what you recorded.

WEEK 6 10m Natural History: Grade 1 - Lesson 30

Baby Birdwatching

 Materials: Walks with a Naturalist 1

PREP: Read Teacher Tip

→ INTRO

Remind me what we read about last week. Look at the picture on p.102. What do you notice? Some birds build nests like this inside those tree holes, so let's peek inside with Melissa and see what's going on in there!

→ READ, NARRATE, & DISCUSS

p.102-105 "In our yard" - "and I did."

• What kinds of materials do you think birds make their nests from? If you were a bird, what would you make your nest from?

→ SUPPLEMENTAL

∞ Video Link: Nest Box Time Lapse (10:25). This video is longer than usual; watch as much or as little as you want or use the chapters to watch portions, as desired.

∞ Video Link: 3 Bluebirds of North America (5:34)

★ TEACHER TIP

It can sometimes be unnerving for students not to get an immediate answer to a question. Encourage them to sit with big questions, like the one in today's suggested discussion, while they continue to make observations.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 6 15m General Science: Grade 1 - Lesson 30

Wildflower Watching Activity

 Materials: Nature Journal

PREP: Read Teacher Tip.

→ INTRO

Today we will spend about 10 minutes noticing which wildflowers are growing in your yard this season.

→ OBSERVE

What colors and shapes do you notice? Where are they growing? Describe the land and any plants or animals you notice around them. Compare any wildflowers to those growing in the garden. In what ways are they similar or different?

→ NARRATE & DISCUSS

Use a Nature Journal to record what the student noticed.

• Do you notice anyone already familiar to you? Do you notice anyone new?

★ TEACHER TIP

Flower observation is always a great time for an object lesson!

• NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.

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WEEK 7 10m Natural History: Grade 1 - Lesson 31

Spider Drums

 Materials: Walks with a Naturalist 1

→ INTRO

Tell me who we met last week. Can you remember back to the spiders we've read about? (garden/orb weaver, jumping spider) Let's meet another spider this week because spiders are busy in the spring, too.

→ READ, NARRATE, & DISCUSS

p.106-107 "One spring" - "that sound?"

- What makes spiders different from the insects, such as the tree hoppers and ants? (Help students remember that insects have 3 body segments and 6 legs, while spiders have 2 body segments and 8 legs.)

→ SUPPLEMENTAL

∞ Video Link: Mother Wolf Spider (0:50)

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 7 15m General Science: Grade 1 - Lesson 31

Meeting an 8-Legged Creature

 Materials: A House Spider's Life

PREP: Read Teacher Tip.

→ INTRO

Look at the picture on the cover. What kinds of spiders have you noticed before? What did they look like? Where were they? Did they have a web?

→ READ, NARRATE, & DISCUSS

A House Spider's Life (whole book)

- The next time you notice a spider, what part of the spider or web do you think you might want to look at more closely?

★ TEACHER TIP

If you are blessed with a spider in a convenient location, take the opportunity for an object lesson and observe it regularly.

• NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.

WEEK 8 10m Natural History: Grade 1 - Lesson 32

Year of the Cicadas

 Materials: Walks with a Naturalist 1

→ INTRO

Tell me who we met last week. Let's meet a very strange insect that we could observe in the spring. Look at the picture on p.108. What do you think this insect is doing? What do you notice?

→ READ, NARRATE, & DISCUSS

p.108-110 "I was born" - "also amazing?"

- Do you know if you have annual or periodical cicadas where you live? Have you noticed their exuvia on your walks?
- How is the cicada different from the wolf spider? (hint: insect vs spider)

→ SUPPLEMENTAL

∞ Video Link: Billions of Cicadas (2:24)

∞ Video Link: How does the annual cicada call? (3:09)

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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WEEK 8 15m General Science: Grade 1- Lesson 32

Noticing Spiders Activity

 Materials: Nature Journal

PREP: Read Teacher Tip.

→ INTRO

Look for spiders this week. They may be active in grass or wooded areas or in garages or basements where they overwintered. If you don't come across a spider, then just observe whatever nature has provided today!

→ OBSERVE

Is this a spider that you have noticed before? How is it similar or different to the one in your book? How many body segments do you see? How many legs does it have? How many eyes does it have? What colors, markings, or textures do you see? Does it have a web? If so, describe the shape and construction of the web.

→ NARRATE & DISCUSS

Use a Nature Journal to record what the student noticed.

- Tell about what you found or what you recorded.

★ TEACHER TIP

If you know where a spider has built a web, do an object lesson! If students struggle to find them, it's okay. Just encourage them to pursue the nature mystery - if the spiders aren't where we have looked, where could they be? Why aren't they here?

- NATURE NOTEBOOK Prompt for General Science in Outdoor Work Quick Link.

WEEK 9 10m Natural History: Grade 1- Lesson 33

Blue Ghosts

 Materials: Walks with a Naturalist 1

→ INTRO

Tell me what you remember about cicadas. Look at the picture on p.112. What do you notice about this insect? This insect has a special trick that is very different from the cicada's.

→ READ, NARRATE, & DISCUSS

p.112-114 "One summer - "near you."

- How is the blue ghost similar to the cicada (and the other insects we've met)? How is it different?

→ SUPPLEMENTAL

- ∞ Video Link: Fireflies in North Carolina (7:03)
- ∞ Video Link: Think there aren't fireflies in the west? (4:00)
- ∞ Video Link: Catching and Chasing Fireflies (3:04)

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

★ TEACHER TIP

If students catch fireflies during their outdoor time, this is a great opportunity for an object lesson.

WEEK 9 15m General Science: Grade 1- Lesson 33

Bird Watching Activity

 Materials: Nature Journal

PREP: Read Teacher Tip.

→ INTRO

Take some time to notice what birds are visiting today.

→ OBSERVE

Spend about 10 minutes bird watching. What colors, markings, and shapes do you notice? What are they doing?

★ TEACHER TIP

If birds are visiting a feeder, use the opportunity to do an object lesson.



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→ NARRATE & DISCUSS

Use a Nature Journal to record what the student noticed.

- Do you notice anyone already familiar to you? Do you notice anyone new?

WEEK 10 10m Natural History: Grade 1 - Lesson 34

Sneaky Spider

Materials: Walks with a Naturalist 1

→ INTRO

Tell me about the insects we've been reading about. Do you recall who we met before them? Today, we meet another spider.

→ READ, NARRATE, & DISCUSS

p.115-117 "Out in my - "catch their prey!"

- Have you ever noticed a spider or other animal that is good at camouflage? If so, tell about it.

→ SUPPLEMENTAL

∞ Video Link: Goldenrod Crab Spider (0:20)

∞ Video Link: Stealth Crab Spider (2:13)

• OUTDOOR WORK

Weekly walks and daily

outdoor time are vital!

Resources in Quick Links.

WEEK 10 15m General Science: Grade 1 - Lesson 34

Wildflower Watching Activity

Materials: Nature Journal

→ INTRO

Today we will spend about 10 minutes noticing which wildflowers are growing in your yard this season.

→ OBSERVE

How many different colors can you find today? Describe the flowers, as well as any plant or critter neighbors. Compare the wildflower to any flowers already growing in the garden.

→ NARRATE & DISCUSS

Use a Nature Journal to record what the student noticed.

- Do you notice anyone already familiar to you? Do you notice anyone new?

WEEK 11 10m Natural History: Grade 1 - Lesson 35

Catch Up

→ CATCH UP

Use this time to catch up on lessons, watch some videos that you didn't have time for, or discuss some interesting ideas over a snack. Depending on your discussions, these MIGHT include:

- What signs of spring (opening buds, baby animals, wildflowers, spiders, insects, etc) have you noticed?
- Tell about surprising places where animals live, such as rotting logs and tree holes.
- Tell about interesting spiders and insects you have seen in the garden.
- Compare and contrast spiders and insects.

• OUTDOOR WORK

Weekly walks and daily

outdoor time are vital!

Resources in Quick Links.



Term 3

WEEK 11 15m General Science: Grade 1 - Lesson 35 Noticing Spiders Activity Materials: Nature Journal → INTRO Look for spiders again this week. You could revisit a spider that you have already met, or maybe meet someone new! → OBSERVE Describe the colors, markings, and surroundings of any spiders you notice. → NARRATE & DISCUSS Use a Nature Journal to record what the student noticed. Tell about what you found or what you recorded.	• NATURE NOTEBOOK Prompt for General Science in Outdoor Work Quick Link.
WEEK 12 10m Natural History: Grade 1 - Lesson 36 Exam Week → EXAMS Answer question(s) related to course. Questions will come from: Walks with a Naturalist 1	• OUTDOOR WORK Weekly walks and daily outdoor time are vital! Resources in Quick Links.
WEEK 12 15m General Science: Grade 1 - Lesson 36 Exam Week → EXAMS Answer question(s) related to course. Questions will come from: Up in the Garden and Down in the Dirt	

Science: Grade 1

Examination

Term 1

General Science: Grade 1

- Over and Under the Pond: Tell about or draw a pond habitat with five things that live there.

Natural History: Grade 1

- Walks with a Naturalist 1: Tell about or draw the ant or the bee. OR Tell about or draw the garden or jumping spider.

Term 2

General Science: Grade 1

- A Fruit is a Suitcase for Seeds: Tell about or draw three seeds that you looked at this term.

Natural History: Grade 1

- Walks with a Naturalist 1: Tell about or draw a plant seed and explain how it moves to find a new home. OR Tell about or draw a bird that you have learned about.

Term 3

General Science: Grade 1

- Up in the Garden and Down in the Dirt: Tell about or draw a garden habitat, including at least five creatures that live in it.

Natural History: Grade 1

- Tell how to roll a log and what you might find underneath it. OR Tell about or draw how spiders are different from insects.